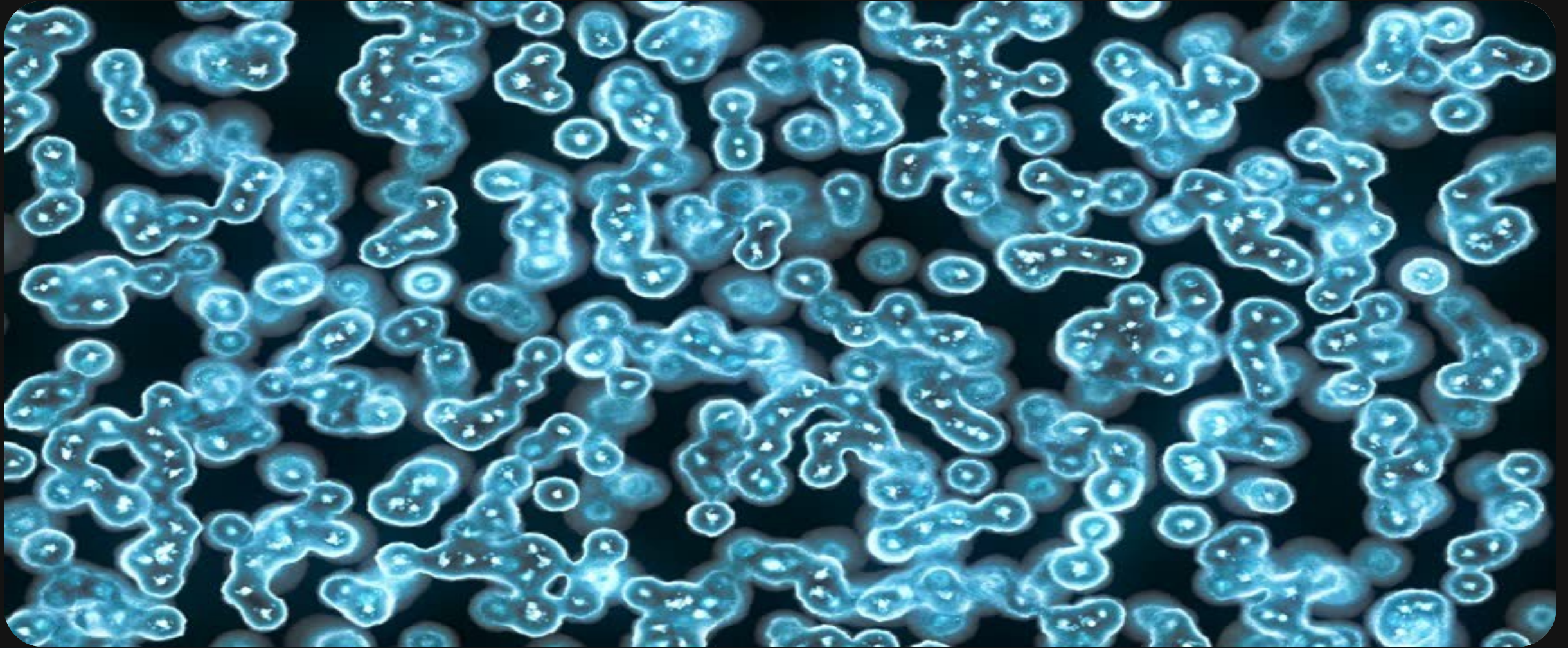
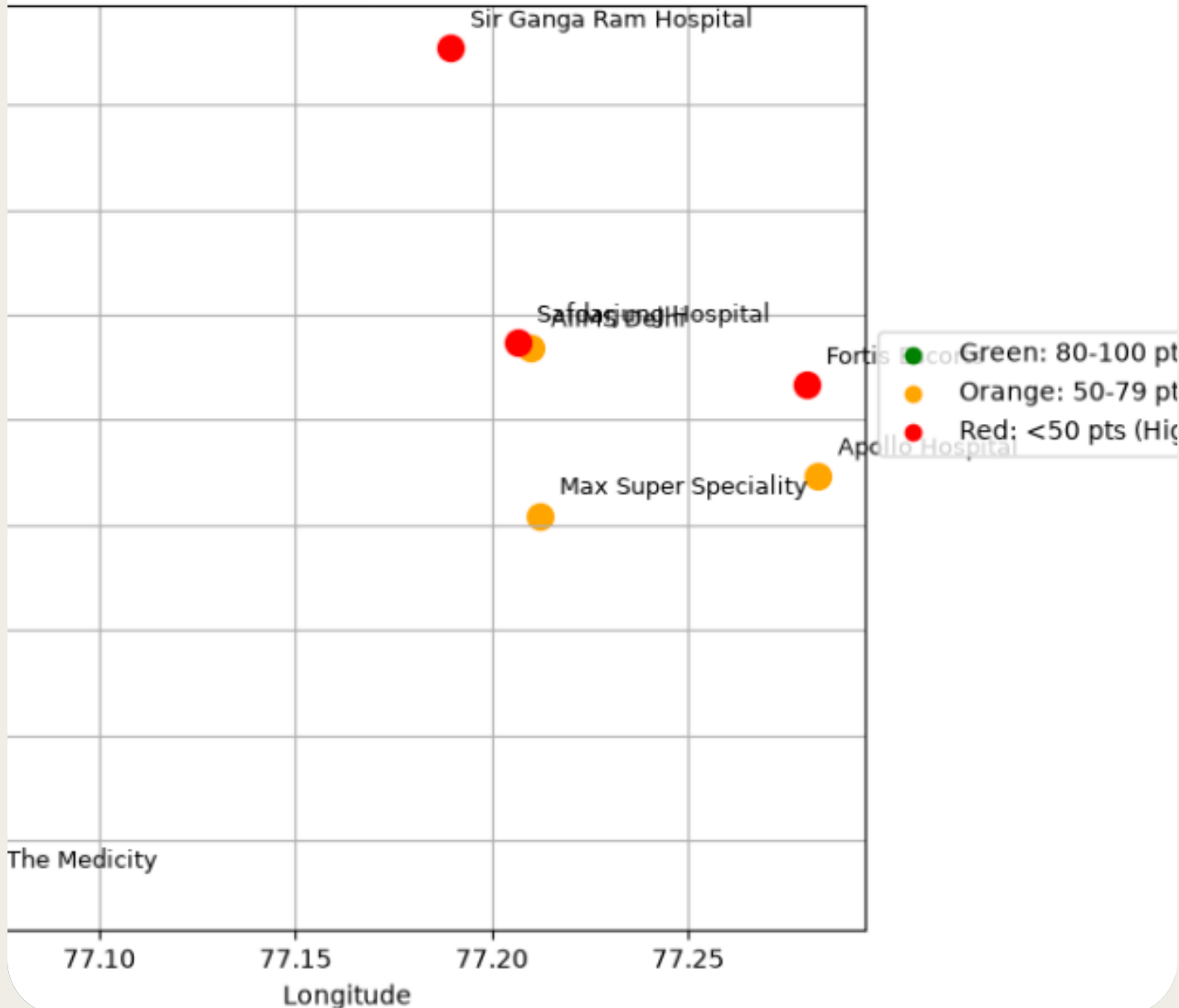




Antimicrobial risk monitoring dashboard

Presented by Neer Nimish Rustagi, Atharva Vrat Chandan and Atharv Tyagi

Delhi NCR Hospital Map & AMR Risk Rating



MAP FUNCTION

The map is calculated after each patient's prescription logging by the appropriate authorities. Based on the prescription, and if the pathogen was identified and the drug was changed, (only if the drug needed to be specialized), an AMR rating is established from 0-100.

Hospital List

Hospital Name	Latitude	Longitude	AMR Score
AIIMS Delhi	28.5672	77.21	71.42857142857143
Safdarjung Hospital	28.5683	77.2064	65.0
Max Super Speciality	28.5273	77.2122	55.55555555555556
Apollo Hospital	28.5367	77.2831	53.84615384615385
Medanta The Medicity	28.4384	77.0433	52.0
Fortis Escorts	28.5583	77.2801	50.0
Sir Ganga Ram Hospital	28.6385	77.1895	59.09090909090909

List of AMR scores of different hospitals, based on 200 randomly generated prescriptions, from 50 different patients.

Patient Registration System

this patient registration system is based on SQL using the sqlite3 library in python.
This is so that it can be readily integrated with any public healthcare system, eg: CGHS and its unique patient keys

Register Patient

[Dashboard](#) [Register Patient](#) [Submit Prescription](#) [View Data](#)

Name

Age

Gender

Male

Register

Prescription processing and data viewing

View Data

Dashboard

Register Patient

Submit Prescription

View Data

ID	Hospital	Patient	Age	Days	Init Drug	Diagnosed?	Condition	Final Drug
200	Apollo Hospital	Patient 29	39	13	Cephalexin	Yes	Sepsis (Various)	Pretomanid (MDR-T
199	Apollo Hospital	Patient 43	33	13	Doxycycline	Yes	Sepsis (Various)	Penicillin
198	AIIMS Delhi	Patient 10	6	7	Ciprofloxacin	No	-	Ciprofloxacin
197	Sir Ganga Ram Hospital	Patient 34	74	12	Clindamycin	Yes	Pneumonia (S. pneumoniae, K. pneumoniae)	-
196	Safdarjung Hospital	Patient 43	33	9	Clindamycin	Yes	Other	Vancomycin (MRSA,
195	Max Super Speciality	Patient 43	33	6	Azithromycin	Yes	Tuberculosis (M. tuberculosis)	Cephalexin
194	Sir Ganga Ram Hospital	Patient 25	33	8	Amoxicillin	Yes	Sepsis (Various)	Azithromycin
193	Apollo Hospital	Patient 19	9	12	Sulfamethoxazole/Trimethoprim	Yes	Throat Infection (Streptococcus pyogenes)	-

Submit Prescription

Dashboard

Register Patient

Submit Prescription

Patient

Patient 1 (Age: 13)

Hospital

AIIMS Delhi

Prescription Count (Days)

Initial Drug

-- Select Common Antibiotic --

Did they diagnose your condition?

Yes

Condition & Pathogen (If diagnosed)

Unknown / None

Final Drug (After Diagnosis)

None / Same as Initial

Submit Prescription

Data can be viewed by database administrators, but names are redacted by default to protect patient security

BIBLIOGRAPHY

1. The websites skeleton was taken from modular opensource website builder Grav ([hyperlink](#))
 - Only skeleton because Grav is for node.js and we coded it in python using flask
2. The code is stored in the GitHub repository <https://github.com/neernimish/PS2iitdsubmission>
3. The longitudes and latitudes for the hospitals were taken from google earth ([hyperlink](#))
4. The code for the random generation is in the file generate_sample.py (run only once)
5. Code for the map was inspired by the Stackoverflow post: ([hyperlink](#))_
[python - Plotting a 2D heatmap - Stack Overflow](#)
4. Libraries needed:
 - Flask
 - Matplotlib
 - Pandas
 - Sqlite (pip install sqlite3)